Nmap Field Reference

Authorized Inventory, Validation, NSE, Output, and Defensive Workflows

Authorized inventory, validation, change detection, and defensive investigation workflows for operators who already know TCP/IP, Unix-like shells, and ordinary Nmap usage.

SCOPE Use only on systems you own, administer, have explicit written permission to test, or operate in a lab. Notify monitoring teams when appropriate, respect scan windows and rate limits, and save evidence without overclaiming.

WARN This reference intentionally omits stealth/evasion playbooks, credential brute force, exploitation, post-exploitation, persistence, malware, and unauthorized access workflows. NSE examples favor default, safe, discovery, version, configuration, and defensive validation scripts.

Operating Defaults

Decision	Default	Escalate only when
Scope control	Use `-iL scope/targets.txt`, `--excludefile scope/exclude_fragile.txt`, and a saved `-sL -n` preflight.	A one-off host check is genuinely safer than editing the shared scope file.
Discovery	Start with `nmap -sn -n`, then add ICMP/TCP/UDP probes for filtered routed networks.	You need narrow `-Pn` confirmation on known approved targets.
TCP inventory	Run SYN top ports first, then full `-p-` only on confirmed live hosts.	Lab, maintenance window, or explicit approval supports deeper coverage.
UDP	Sample selected UDP ports and top UDP ports.	A service owner asks for protocol-specific validation.
NSE	Prefer named scripts or `"default and safe"` / `"safe and discovery"`.	A reviewed script and narrow target/port are approved.
Output	Use `-oA outputs/DATE_SCOPE_NAME` and parse XML for automation.	A quick console check is explicitly disposable.
Reporting	Say "candidate finding" until validated by config, owner, vendor, or packet evidence.	Evidence is confirmed and remediation owner agrees on impact.

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Command Pattern Index

Task	Command pattern	Use and notes
Host discovery	`nmap -sn -n -iL scope/targets.txt -oA outputs/discovery_YYYYMMDD_SCOPE`	Fast live-host pass. Use `-n` to avoid DNS delays and reverse lookup noise.
Local LAN discovery	`sudo nmap -sn 192.0.2.0/24 -oA outputs/lan_discovery`	Root/admin enables ARP discovery on local Ethernet, usually more reliable than routed probes.
Routed discovery	`nmap -sn -PE -PS22,80,443 -PA80,443 -PU53,123 -iL scope/targets.txt -oA outputs/routed_discovery`	Mix ICMP, TCP, and selected UDP probes for routed networks.
Port discovery	`sudo nmap -sS --top-ports 1000 --open --reason -iL live-hosts.txt -oA outputs/tcp_top`	Root/admin SYN scan, open-only view, reason codes for review.
Full TCP inventory	`sudo nmap -sS -p- --open --reason -iL live-hosts.txt -oA outputs/tcp_all`	Find all TCP listeners before deeper service detection.
Service/version detection	`nmap -sV --version-light -p T:22,80,443,8080 -iL live-hosts.txt -oA outputs/sv_light`	Light probes first, then raise intensity only where needed.
OS fingerprinting	`sudo nmap -O --osscan-limit -iL live-hosts.txt -oA outputs/os_clues`	Treat as a clue. Needs raw packet privileges and suitable open/closed port mix.
UDP sampling	`sudo nmap -sU --top-ports 50 --open --reason -iL live-hosts.txt -oA outputs/udp_top50`	Slow and ambiguous. Start small and expect rate limiting.
IPv6 scanning	`nmap -6 -sn -iL scope/ipv6_targets.txt -oA outputs/ipv6_discovery`	IPv6 target list should be explicit and authorized.
Script scanning	`nmap -sV --script "default and safe" -p 22,80,443 TARGET -oA outputs/default_safe`	Restricts default NSE to scripts also categorized safe.
Safe vulnerability checks	`nmap -sV --script ssl-ccs-injection -p 443 TARGET -oA outputs/tls_ccs_check`	Run one reviewed script by name against a narrow target/port.
Output formats	`nmap -sV -oA outputs/survey_YYYYMMDD_SCOPE -iL live-hosts.txt`	`-oA` writes normal, XML, and grepable outputs with one basename.
Diffing results	`ndiff outputs/last.xml outputs/current.xml > reports/ndiff_YYYYMMDD.txt`	Use stable scan options, target lists, and timing for meaningful deltas.
Inventory generation	`xmlstarlet sel -T -t -m '//host/ports/port[state/@state="open"]' -v 'ancestor::host/address[1]/@addr' -o 't' -v '@protocol' -o 't' -v '@portid' -o 't' -v 'service/@name' -n outputs/survey.xml > reports/services.tsv`	XML is the automation source of truth.
Subnet validation	`nmap -sL -n -iL scope/targets.txt --excludefile scope/exclude.txt -oN reports/preflight_targets.txt`	List scan before probing. Confirms expansion and exclusions.
Web enumeration	`nmap -sV --script http-title,http-headers,http-methods -p 80,443,8080,8443 TARGET -oA outputs/web_enum`	Discovery-focused HTTP metadata. Validate virtual hosts separately.
TLS checks	`nmap -sV --script ssl-cert,ssl-enum-ciphers -p 443,8443 TARGET -oA outputs/tls_inventory`	Certificate and cipher metadata. Can be noisy on large estates.
DNS checks	`nmap -sU -sT -p U:53,T:53 --script dns-recursion,dns-service-discovery TARGET -oA outputs/dns_checks`	Internal DNS validation only; recursion may be expected internally.
SMTP checks	`nmap -sV --script smtp-commands,smtp-ntlm-info -p 25,465,587 TARGET -oA outputs/smtp_checks`	Capability and NTLM disclosure review. Do not run user enumeration or brute scripts by default.
SMB checks	`nmap -sV --script smb2-security-mode,smb2-time,smb-protocols -p 445 TARGET -oA outputs/smb_checks`	Internal Windows network posture. Scope tightly.
SSH checks	`nmap -sV --script ssh-hostkey,ssh2-enum-algos -p 22 TARGET -oA outputs/ssh_inventory`	Host key and algorithm inventory. Useful for crypto hygiene.
RDP checks	`nmap -sV --script rdp-enum-encryption,rdp-ntlm-info -p 3389 TARGET -oA outputs/rdp_checks`	Encryption and NTLM metadata, internal only.
Timing/performance	`sudo nmap -sS --max-rate 200 --max-retries 3 --host-timeout 30m -iL live-hosts.txt -oA outputs/conservative`	Cap rate before increasing speed. Measure packet loss.
False negatives	`nmap --reason -Pn -p 22,443 TARGET -oA outputs/confirm_pn`	Narrow `-Pn` checks can validate filtered discovery, but broad `-Pn` is noisy.
Defensive reporting	`nmap -sV --version-light --reason -oA outputs/evidence_YYYYMMDD_SCOPE -iL scope/targets.txt`	Preserve command, scope, date, Nmap version, XML, and limitations.

Scan Intent Matrix

Intent	Command shape	Privileges	Output	Footguns
Quick live-host discovery	`nmap -sn -n CIDR -oA outputs/discovery`	None for routed; root/admin improves local ARP	XML plus grepable	Firewalls can drop probes; `-n` avoids DNS but loses names.
Full TCP port inventory	`sudo nmap -sS -p- --open --reason -iL live-hosts.txt -oA outputs/tcp_all`	Root/admin for SYN	XML is primary	Full port scans can trigger alerts and stress fragile systems.
Top ports scan	`sudo nmap -sS --top-ports 1000 --open -iL live-hosts.txt -oA outputs/tcp_top`	Root/admin preferred	Normal for review, XML for parsing	Top ports miss custom services and high ports.
Service/version survey	`nmap -sV --version-light -p PORTS -iL live-hosts.txt -oA outputs/sv`	Usually none	XML for inventory	Banners lie. Version probes can tickle fragile services.
UDP service sampling	`sudo nmap -sU --top-ports 50 --open --reason -iL live-hosts.txt -oA outputs/udp_top`	Root/admin	XML plus normal	ICMP rate limiting creates `open|filtered` ambiguity.
Authenticated boundary	`nmap -sV --script "default and safe" -p PORTS TARGET -oA outputs/safe_nse`	Depends on scan type	XML	Nmap is mostly unauthenticated network perspective; confirm config with platform tools.
Change detection over time	`ndiff outputs/week1.xml outputs/week2.xml > reports/delta.txt`	None	Text delta and saved XML	Inconsistent timing, DNS, scope, or packet loss creates false changes.
Internal asset inventory	`nmap -sn -n -iL scope/targets.txt -oA outputs/discovery` then scan live hosts	Root/admin optional	XML, TSV, report	DHCP churn and duplicate hostnames require reconciliation.
Safe NSE audit	`nmap -sV --script "safe and discovery" -p 22,80,443 TARGET -oA outputs/nse_safe`	Usually none	XML and normal	`safe` does not mean zero impact; read script help.
Noisy/deep scan	`sudo nmap -sS -sU -A -p T:1-65535,U:53,123 TARGET -oA outputs/deep_lab`	Root/admin	XML	Lab or explicit approval only. `-A` bundles version, OS, scripts, traceroute.
Fragile network caution	`sudo nmap -sS --max-rate 50 --scan-delay 100ms --max-retries 5 -p 22,80,443 -iL fragile.txt -oA outputs/fragile`	Root/admin for SYN	XML	Coordinate windows and stop conditions; exclude OT, printers, medical, and legacy gear.

FOOTGUN Broad `-Pn`, `-A`, `--script vuln`, UDP scanning, and full TCP `-p-` scans are easy to justify one host at a time and easy to misuse at subnet scale. Stage scans and keep the blast radius visible.

Target Specification and Scope Control

Scope shape	Example	Notes
Single host	`nmap -sV TARGET -oA outputs/one_host`	Use an IP when DNS identity is not part of the test.
Multiple hosts	`nmap -sV 192.0.2.10 192.0.2.20 app01.example.test`	Good for small spot checks.
CIDR range	`nmap -sn -n 192.0.2.0/24 -oA outputs/cidr_discovery`	Confirm CIDR ownership before scanning.
Nmap range syntax	`nmap -sn -n 192.0.2.10-50 -oA outputs/range_discovery`	Useful for lab blocks; avoid accidental wide ranges.
Input list	`nmap -sn -n -iL scope/targets.txt -oA outputs/discovery`	Best default for approved scopes and repeatability.
Exclude inline	`nmap -sn 192.0.2.0/24 --exclude 192.0.2.5,192.0.2.9`	Fine for one-off exclusions.
Exclude file	`nmap -sn -iL scope/targets.txt --excludefile scope/exclude_fragile.txt`	Prefer for printers, OT, fragile, regulated, or out-of-window systems.
IPv6	`nmap -6 -sn -iL scope/ipv6_targets.txt -oA outputs/ipv6_discovery`	Use explicit lists. IPv6 space is too large for naive subnet scanning.
Hostnames	`nmap -sV app01.example.test -oA outputs/app01`	DNS changes can change target identity. Preserve resolved IPs in reports.
No DNS	`nmap -n -sn -iL scope/targets.txt -oA outputs/discovery_nodns`	Faster and avoids resolver side effects.
Force reverse DNS	`nmap -R -sn -iL scope/targets.txt -oA outputs/discovery_rdns`	Useful for inventory, slower and potentially noisy.
Interface selection	`nmap --iflist` then `sudo nmap -e en0 -sn 192.0.2.0/24`	Use when VPN or multiple routes confuse path selection.
Source address note	Verify route and bind behavior before scanning from multihomed hosts.	Avoid source spoofing for routine inventory. Treat `-S` as lab/routing validation only with explicit authorization.

Scope Preflight

```
mkdir -p scope outputs reports
awk '{print $1}' scope/raw_scope.txt | sed '/^#/d;/^$/d' | sort -u > scope/targets.txt
awk '{print $1}' scope/raw_exclusions.txt | sed '/^#/d;/^$/d' | sort -u > scope/exclude_fragile.txt
nmap -sL -n -iL scope/targets.txt --excludefile scope/exclude_fragile.txt -oN reports/preflight_targets.txt
```

Check	What to look for
Target count	Does the list scan expand to the expected number of hosts?
Out-of-scope ranges	Any public, partner, cloud, or production ranges not approved?
Exclusions	Are printers, OT, fragile, and maintenance-freeze hosts excluded?
DNS behavior	If names are required, compare **`-n`** and **`-R`** outputs intentionally.
Interface/route	Use **`nmap --iflist`** and OS route tools before scanning over VPN.

WARN **`-sL`** is a list scan, not a port scan. Without **`-n`**, name resolution can still query DNS. Use **`-n`** for a clean expansion check.

Host Discovery

Mode	Command	Defensive use	Caveats
Default discovery	**`nmap -sn CIDR -oA outputs/discovery_default`**	Let Nmap choose ordinary probes.	Behavior differs by privilege and local vs routed network.
Ping scan only	**`nmap -sn -n -iL scope/targets.txt -oA outputs/discovery`**	Inventory live candidates before deeper scans.	Firewalls cause false negatives.
ARP local LAN	**`sudo nmap -sn 192.0.2.0/24 -oA outputs/arp_lan`**	Local Ethernet discovery, generally reliable.	Only local link. Needs correct interface/VLAN.
ICMP probes	**`nmap -sn -PE -PP -PM -iL scope/targets.txt -oA outputs/icmp_discovery`**	Test echo, timestamp, netmask responses.	Often filtered on routed networks.
TCP SYN discovery	**`sudo nmap -sn -PS22,80,443,3389 -iL scope/targets.txt -oA outputs/ps_discovery`**	Check likely service ports without full scan.	Root/admin sends SYN; unprivileged behavior may differ.
TCP ACK discovery	**`sudo nmap -sn -PA80,443 -iL scope/targets.txt -oA outputs/pa_discovery`**	Find hosts behind permissive stateful paths.	Firewalls may drop ACKs.
UDP discovery	**`sudo nmap -sn -PU53,123,161 -iL scope/targets.txt -oA outputs/pu_discovery`**	Internal DNS/NTP/SNMP liveness hints.	Can be slow; ICMP unreachable rate limiting matters.
Treat as up	**`nmap -Pn -p 22,443 TARGET -oA outputs/confirm_ports`**	Narrow confirmation when discovery is filtered.	Broad **`-Pn`** scans every target as live and can be very noisy.

Discovery to Next Stage

```
nmap -sn -n -iL scope/targets.txt -oA outputs/discovery_YYYYMMDD_SCOPE
awk '/Status: Up/{print $2}' outputs/discovery_YYYYMMDD_SCOPE.gnmap | sort -u > scope/live-hosts.txt
sudo nmap -sS --top-ports 1000 --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_top_YYYYMMDD_SCOPE
```

Situation	Adjustment
No hosts found	Try **`-PE -PS22,80,443 -PA80,443`**, check VPN/interface, then narrow **`-Pn`** against a known approved host.
Local subnet via VPN	Check **`nmap --iflist`**; your default route may not be the scan route.
Mixed Windows/Linux estate	Combine ICMP and TCP discovery. Windows firewalls often drop some ICMP.
Asset has host firewall	Treat discovery as a filter test, not proof the asset is absent.
IDS visibility	Discovery probes are visible. Notify SOC when scanning production.

TCP Port Scanning Strategies

Strategy	Command	When to use	Notes
SYN scan	`sudo nmap -sS --top-ports 1000 --open --reason TARGET -oA outputs/tcp_syn_top`	Default privileged TCP inventory.	Raw packet privileges; fast and flexible.
Connect scan	`nmap -sT -p 22,80,443 TARGET -oA outputs/tcp_connect`	Unprivileged account or constrained host.	Completes TCP connect; can be noisier in logs.
ACK firewall mapping	`sudo nmap -sA -p 80,443 TARGET -oA outputs/firewall_ack`	Defensive path validation and filter mapping.	Does not identify open services; reports filtering behavior.
Top ports	`sudo nmap -sS --top-ports 2000 --open -iL live-hosts.txt -oA outputs/top2000`	Good inventory pass.	Misses uncommon custom ports.
Explicit list	`sudo nmap -sS -p 22,80,443,445,3389 TARGET -oA outputs/selected_tcp`	Focused validation.	Keep port list tied to purpose.
Port range	`sudo nmap -sS -p 1-1024,8000-9000 TARGET -oA outputs/ranges`	Known service bands.	Ranges can expand faster than expected.
All TCP ports	`sudo nmap -sS -p- --open --reason TARGET -oA outputs/all_tcp`	Full listener inventory.	Stage by live hosts and use scan windows.
Fast mode	`nmap -F TARGET -oA outputs/fast`	Quick triage.	Fewer ports than default; not an inventory substitute.
Open-only	`nmap --open -p- TARGET -oA outputs/open_only`	Reduce report noise.	Hidden closed/filtered detail can matter for diagnostics.
Reason output	`nmap --reason -p 22,443 TARGET -oA outputs/reason`	Evidence and troubleshooting.	Adds scan-state rationale.
Packet trace	`nmap --packet-trace -p 443 TARGET -oN reports/packet_trace.txt`	Lab troubleshooting only.	Very verbose; can include sensitive metadata.

NOTE Null, FIN, Xmas, Window, and Maimon scans are historical or edge-case diagnostics. Do not use them as stealth playbooks. Prefer ordinary SYN/connect scans and transparent monitoring coordination.

Staged TCP Inventory

```
sudo nmap -sS -p- --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_all_YYYYMMDD_SCOPE
xmlstarlet sel -T -t -m '//host/ports/port[state/@state="open"]' \
  -v 'ancestor::host/address[1]/@addr' -o ' ' -v '@portid' -n \
  outputs/tcp_all_YYYYMMDD_SCOPE.xml > reports/open_tcp_host_ports.txt
nmap -sV --version-light -iL scope/live-hosts.txt -p T:22,80,443,445,3389 -oA outputs/sv_common_YYYYMMDD_SCOPE
```

Footgun	Control
Full scan on unfiltered production subnet	Discover live hosts first, then scan in batches.
Service detection against every closed/filtered port	Discover open ports first, then target version probes.
Long custom port list in shell history	Keep commands in `commands.log` and protect sensitive outputs.
Assuming `filtered` means secure	It only means Nmap could not determine the port state.

UDP Scanning Strategies

Strategy	Command	Use	Footguns
Top UDP sample	`sudo nmap -sU --top-ports 50 --open --reason TARGET -oA outputs/udp_top50`	First-pass internal service sampling.	Slow; expect `open|filtered`.
Selected UDP	`sudo nmap -sU -p 53,67,68,69,123,161,500,514,1812,1900 TARGET -oA outputs/udp_selected`	Validate specific approved services.	Some ports are chatty or fragile.
UDP with version	`sudo nmap -sU -sV --version-light -p 53,123,161 TARGET -oA outputs/udp_sv`	Add service clues where probes are safe.	Version probes can change timing and load.
TCP plus UDP split	`sudo nmap -sS -p 22,80,443 TARGET -oA outputs/tcp_selected` then UDP separately	Easier impact control.	Avoid `-sS -sU -p-` style blanket scans.
DNS validation	`sudo nmap -sU -sT -p U:53,T:53 --script dns-recursion TARGET -oA outputs/dns_recursion`	Internal resolver posture.	Recursion is context dependent; do not overclaim.
NTP info	`sudo nmap -sU -p 123 --script ntp-info TARGET -oA outputs/ntp_info`	Time service inventory.	Old NTP devices can be fragile.
SNMP info	`sudo nmap -sU -p 161 --script snmp-sysdescr --script-args creds.snmp=COMMUNITY TARGET -oA outputs/snmp_sysdescr`	Only with explicit community authorization.	Community strings are secrets. Protect outputs.
IPsec presence	`sudo nmap -sU -p 500,4500 TARGET -oA outputs/ipsec_ports`	Detect IKE/NAT-T listeners.	Detection only; do not brute or negotiate aggressively.
Kerberos/LDAP sample	`sudo nmap -sU -sT -p U:88,389,T:88,389 TARGET -oA outputs/directory_ports`	Directory service exposure review.	Domain controllers are sensitive; narrow windows.

WARN UDP ambiguity is normal. A UDP service may be open but silent, filtered by path devices, or hidden behind ICMP rate limits. Report UDP findings with confidence language.

Service and Version Detection

Option	Command	Use	Notes
Basic version detection	`nmap -sV -p 22,80,443 TARGET -oA outputs/sv_basic`	Identify service names and versions.	Banners can be false, stale, or customized.
Light probes	`nmap -sV --version-light -p PORTS TARGET -oA outputs/sv_light`	Lower-impact first pass.	May miss uncommon services.
Intensity	`nmap -sV --version-intensity 5 -p PORTS TARGET -oA outputs/sv_i5`	Tune probe aggressiveness.	Range is 0 to 9. Higher is more probing.
All probes	`nmap -sV --version-all -p PORTS TARGET -oA outputs/sv_all`	Lab or narrow unresolved services.	Explicit approval for sensitive systems.
Trace probes	`nmap -sV --version-trace -p 443 TARGET -oN reports/version_trace.txt`	Troubleshoot odd service detection.	Very verbose; use one host/port.
CPE output	`nmap -sV -p PORTS TARGET -oX outputs/sv.xml`	Extract CPEs for inventory.	CPEs are clues, not patch proof.
Aggressive bundle	`sudo nmap -A TARGET -oA outputs/aggressive_lab`	Lab or explicit deep validation.	**-A** includes OS detection, version detection, scripts, and traceroute.

Extract Service Data

```
xmlstarlet sel -T -t \
  -m '//host/ports/port[state/@state="open"]' \
  -v 'ancestor::host/address[1]/@addr' -o '\t' \
  -v '@protocol' -o '\t' -v '@portid' -o '\t' \
  -v 'service/@name' -o '\t' \
  -v 'service/@product' -o '\t' \
  -v 'service/@version' -o '\t' \
  -v 'service/cpe[1]' -n outputs/sv.xml > reports/services.tsv
```

Interpretation rule	Field practice
Banner says Apache 2.4.x	Candidate evidence; verify package manager, container image, or config.
Product blank but service known	Keep the port/service in inventory and schedule targeted validation.
TLS-wrapped service	Use **-sV** plus TLS scripts on the TLS port; cleartext assumptions fail.
Service tunnel/proxy	Record front-end and back-end uncertainty.
CPE present	Useful for enrichment; not a vulnerability finding by itself.

OS Detection and Network Fingerprinting

Option	Command	Use	Caveats
OS detection	`sudo nmap -O TARGET -oA outputs/os_detect`	OS family clue.	Probabilistic. Needs raw packet privileges.
Limit to promising targets	`sudo nmap -O --osscan-limit -iL live-hosts.txt -oA outputs/os_limited`	Avoid wasting probes where prerequisites are absent.	Best when target has open and closed TCP ports.
More guessing	`sudo nmap -O --osscan-guess TARGET -oA outputs/os_guess`	Lab or stubborn target.	Lower confidence; label clearly.
Bundle caution	`sudo nmap -A TARGET -oA outputs/aggressive_lab`	Lab deep profile.	Adds scripts and traceroute; not just OS detection.

Factor	Effect
NAT/load balancer	Fingerprint may describe the intermediary, not the host.
Host firewall	Alters port states and TCP behavior.
Virtualization	Many guests share similar network fingerprints.
Middleboxes	TTL, window, and reset behavior can reflect the path.
No closed port	OS detection quality often drops.
No open port	OS detection quality often drops.

NOTE Report OS results as "Nmap OS guess" with confidence. Do not treat fingerprint output as authoritative asset classification.

Timing, Performance, and Reliability

Control	Example	Use	Risk
Timing template	`nmap -T3 -sS --top-ports 1000 TARGET`	Practical default for normal networks.	Templates combine several timing choices.
Conservative	`sudo nmap -sS --max-rate 100 --max-retries 5 --scan-delay 50ms TARGET`	Fragile, congested, or monitored paths.	Slower, but easier to explain.
Faster lab	`sudo nmap -sS --min-rate 5000 --max-retries 2 -p- TARGET`	Controlled lab where packet loss is understood.	Can overwhelm networks outside labs.
Cap rate	`--max-rate 200`	Put a ceiling on probe rate.	Too low can make scans impractical.
Minimum rate	`--min-rate 1000`	Force speed for lab scans.	Can increase packet loss and false negatives.
Retries	`--max-retries 3`	Bound scan duration.	Lower retries reduce confidence on lossy paths.
Host timeout	`--host-timeout 30m`	Prevent one host from stalling a batch.	Can skip slow but important hosts.
Scan delay	`--scan-delay 100ms --max-scan-delay 1s`	Slow probes for fragile services or rate-limited paths.	Extends duration.
Parallelism	`--min-parallelism 10 --max-parallelism 100`	Advanced tuning after measuring behavior.	Over-tuning creates confusing results.
RST rate-limit note	`--defeat-rst-ratelimit`	Troubleshoot closed-port RST limiting in approved tests.	Do not use as stealth/evasion guidance.

Profiles

```
# Conservative internal scan
sudo nmap -sS --top-ports 1000 --open --reason \
  --max-rate 200 --max-retries 5 --host-timeout 30m \
  -iL scope/live-hosts.txt -oA outputs/tcp_conservative_YYYYMMDD_SCOPE

# Faster lab scan
sudo nmap -sS -p- --open --reason --min-rate 3000 --max-retries 2 \
  -iL scope/lab-hosts.txt -oA outputs/lab_alltcp_YYYYMMDD

# Resume-safe large scan
sudo nmap -sS -p- --open --reason --max-retries 3 --host-timeout 45m \
  -iL scope/live-hosts.txt -oA outputs/alltcp_YYYYMMDD_SCOPE
nmap --resume outputs/alltcp_YYYYMMDD_SCOPE.nmap
```

Reliability issue	Defensive response
Packet loss	Lower rate, increase retries, scan smaller batches, compare with path telemetry.
IDS/IPS alerts	Pre-notify, tag source IP, save commands, and correlate timestamps.
Fragile devices	Exclude by file, scan only required ports, set stop condition.
Maintenance window	Use `--host-timeout`, batch files, and resume.
False negatives	Repeat a small sample with slower timing before reporting absence.

Output Formats and Reporting

Output	Flag	Use	Notes
Normal	`-oN outputs/scan.nmap`	Human review and resume source.	Do not parse it for automation.
XML	`-oX outputs/scan.xml`	Automation, inventory, reports, `ndiff`.	Best machine source.
Grepable	`-oG outputs/scan.gnmap`	Legacy quick shell extraction.	Deprecated but still useful.
All major formats	`-oA outputs/scan`	Writes `.nmap`, `.xml`, `.gnmap`.	Recommended default.
Append	`--append-output -oN outputs/log.nmap`	Long manual sessions.	Avoid mixing unrelated scans.
Resume	`nmap --resume outputs/scan.nmap`	Continue interrupted scan.	Do not change options by hand.
Stylesheet	`--stylesheet path/to/nmap.xsl`	XML-to-HTML workflows.	Keep local copy for reproducibility.
Web XML	`--webxml`	Portable reference to Nmap stylesheet.	External fetch may be blocked.
Verbosity	`-v`, `-vv`	More progress and detail.	Can inflate output.
Debug	`-d`, `-dd`	Troubleshooting.	Noisy; save separately.

Project Output Pattern

```
STAMP=$(date -u +%Y%m%dT%H%M%SZ)
SCOPE=corp-lab
mkdir -p scope outputs reports
nmap -sV --version-light --reason -iL scope/live-hosts.txt -oA "outputs/${STAMP}_${SCOPE}_sv"
```

XML to HTML or TSV

```
# Convert XML to HTML when xsltproc and a local stylesheet are available.
xsltproc /opt/homebrew/share/nmap/nmap.xsl outputs/scan.xml > reports/scan.html

# Extract open host, protocol, port, service.
xmlstarlet sel -T -t \
  -m '//host/ports/port[state/@state="open"]' \
  -v 'ancestor::host/address[1]/@addr' -o '\t' \
  -v '@protocol' -o '\t' -v '@portid' -o '\t' \
  -v 'service/@name' -n outputs/scan.xml > reports/open_services.tsv
```

FOOTGUN Human output is for humans. Normal output changes are harder to parse safely than XML. Build automation on XML, not screen text.

NSE: Nmap Scripting Engine

Topic	Command or rule	Defensive note
Default scripts	**`nmap -sC -p 22,80,443 TARGET -oA outputs/default_scripts`**	**`-sC`** equals **`--script=default`**; use with permission.
Safe default subset	**`nmap -sV --script "default and safe" -p PORTS TARGET -oA outputs/default_safe`**	Good baseline when you want default-style output with safety filter.
Discovery scripts	**`nmap -sV --script "safe and discovery" -p PORTS TARGET -oA outputs/discovery_scripts`**	Still read help and scope carefully.
Single script	**`nmap -sV --script ssl-cert -p 443 TARGET -oA outputs/ssl_cert`**	Prefer named scripts for repeatable evidence.
Script args	**`nmap --script snmp-sysdescr --script-args creds.snmp=COMMUNITY -sU -p 161 TARGET`**	Treat args as secrets when they contain credentials or communities.
Script help	**`nmap --script-help ssl-cert`**	Read before running unfamiliar scripts.
Locate scripts	**`nmap --script-help http-title`** or inspect local scripts directory	Script names vary by install/version.
Update database	**`sudo nmap --script-updatedb`**	Needed after installing local NSE scripts.
Script output	Save **`-oA`**	Script text is useful; XML preserves structure better.
Trace script traffic	**`--script-trace`**	Lab/debug only; can expose sensitive data.

Category	Default stance
`default`	Reasonable baseline, but read scope and impact.
`safe`	Preferred for routine discovery, still not risk-free.
`discovery`	Good for inventory; can be chatty.
`version`	Used by version detection; not selected directly like ordinary categories.
`vuln`	Narrow validation only; read help and run by name where possible.
`auth`	Avoid unless the purpose is explicitly authorized access/config validation.
`brute`	Avoid by default. Not an inventory technique.
`exploit`	Avoid outside explicitly approved lab validation.
`intrusive`	Avoid by default; can be disruptive.
`dos`	Avoid. Can intentionally or accidentally disrupt services.
`external`	Privacy review required; may send target data to third parties.

Defensive NSE Examples

```
nmap --script-help ssl-cert
nmap -sV --script "default and safe" -p 22,80,443 TARGET -oA outputs/default_safe_TARGET
nmap -sV --script http-title,http-headers,http-methods -p 80,443,8080,8443 TARGET -oA outputs/http_meta_TARGET
nmap -sV --script ssl-cert,ssl-enum-ciphers -p 443,8443 TARGET -oA outputs/tls_TARGET
nmap -sV --script ssh-hostkey,ssh2-enum-algos -p 22 TARGET -oA outputs/ssh_crypto_TARGET
nmap -sV --script smb2-security-mode,smb2-time,smb-protocols -p 445 TARGET -oA outputs/smb_posture_TARGET
nmap -sU -sT -p U:53,T:53 --script dns-recursion,dns-service-discovery TARGET -oA outputs/dns_TARGET
```

WARN Do not run **`--script all`**, **`brute`**, **`exploit`**, **`dos`**, or broad **`vuln`** categories as a default workflow. If a script can change state, consume significant resources, disclose data externally, or test exploitation, it needs explicit authorization and a narrow plan.

Defensive Vulnerability Validation With NSE

Practice	Command	Reporting language
List available vuln scripts	**`find /opt/homebrew/share/nmap/scripts -name '*vuln*.nse' -maxdepth 1`**	Local script database, not a universal catalog.
Read script help	**`nmap --script-help smb-vuln-ms17-010`**	Record categories and script behavior.
Run one safe check	**`nmap -sV --script smb-vuln-ms17-010 -p 445 TARGET -oA outputs/ms17_010_TARGET`**	"Nmap script reported candidate MS17-010 exposure; confirm patch state."
TLS vuln check	**`nmap -sV --script ssl-ccs-injection -p 443 TARGET -oA outputs/ssl_ccs_TARGET`**	Candidate finding; validate with vendor-supported tooling.
Capture audit trail	**`nmap -sV --script SCRIPT -p PORT TARGET -oA outputs/validation_YYYYMMDD_TARGET`**	Preserve XML, command, script help, and Nmap version.
Avoid broad category	Prefer **`--script SCRIPTNAME`** over **`--script vuln`**	Reduces surprise impact and false positives.

Guardrail	Why it matters
Read help first	Some **`vuln`** scripts are safe checks; others are intrusive or protocol-specific.
Narrow target and port	Reduces scan impact and improves evidence quality.
Use lab/safe window	Validation traffic may resemble attack traffic to monitoring.
Verify externally	Vendor tools, configuration review, package versions, and compensating controls decide final status.
Do not overclaim	Script output is evidence, not proof by itself.

Protocol-Focused Recipes

Protocol	Command	Checks	Save	Footguns and narrowing
HTTP/HTTPS	`nmap -sV --script http-title,http-headers,http-methods -p 80,443,8080,8443 TARGET -oA outputs/http_TARGET`	Titles, headers, allowed methods, service guess.	XML and normal.	Virtual hosts may hide apps; Nmap is not a web scanner.
TLS certs/ciphers	`nmap -sV --script ssl-cert,ssl-enum-ciphers -p 443,8443 TARGET -oA outputs/tls_TARGET`	Certificate identity, validity, cipher/protocol support.	XML for inventory.	Cipher enumeration can be slower; batch carefully.
SSH	`nmap -sV --script ssh-hostkey,ssh2-enum-algos -p 22 TARGET -oA outputs/ssh_TARGET`	Host keys, algorithms, version.	XML plus normal.	Do not run brute or auth tests by default.
DNS	`nmap -sU -sT -p U:53,T:53 --script dns-recursion,dns-service-discovery TARGET -oA outputs/dns_TARGET`	DNS service presence, recursion, DNS-SD.	XML.	Internal recursion may be intended; external recursion is context.
SMTP	`nmap -sV --script smtp-commands,smtp-ntlm-info -p 25,465,587 TARGET -oA outputs/smtp_TARGET`	Capabilities and NTLM metadata.	XML.	Avoid open-relay or user-enum checks unless explicitly approved.
SMB	`nmap -sV --script smb-protocols,smb2-security-mode,smb2-time -p 445 TARGET -oA outputs/smb_TARGET`	SMB dialects, signing mode, time.	XML.	Domain controllers and file servers are sensitive; narrow scope.
RDP	`nmap -sV --script rdp-enum-encryption,rdp-ntlm-info -p 3389 TARGET -oA outputs/rdp_TARGET`	Encryption and NTLM info.	XML.	Can trigger security logs.
LDAP	`nmap -sV --script ldap-rootdse -p 389,636 TARGET -oA outputs/ldap_TARGET`	RootDSE naming and capability metadata.	XML.	Directory servers are high value; do not enumerate broadly without approval.
Kerberos	`nmap -sT -sU -p T:88,U:88 TARGET -oA outputs/kerberos_ports`	Service exposure only.	XML.	Avoid user enumeration and auth guessing.
SNMP	`sudo nmap -sU -p 161 --script snmp-sysdescr --script-args creds.snmp=COMMUNITY TARGET -oA outputs/snmp_TARGET`	System info with approved community.	Protect XML.	Community string is sensitive; do not guess.
NTP	`sudo nmap -sU -p 123 --script ntp-info TARGET -oA outputs/ntp_TARGET`	NTP metadata.	XML.	Avoid old monlist-style broad checks unless explicitly required.
PostgreSQL	`nmap -sV -p 5432 --script pgsql-info TARGET -oA outputs/pgsql_TARGET`	Version and protocol info.	XML.	Do not brute or query data.
MySQL	`nmap -sV -p 3306 --script mysql-info TARGET -oA outputs/mysql_TARGET`	Version and capability info.	XML.	Avoid empty password/brute scripts by default.
MSSQL	`nmap -sV -p 1433 --script ms-sql-info,ms-sql-ntlm-info TARGET -oA outputs/mssql_TARGET`	Instance/version and NTLM metadata.	XML.	Avoid DAC/query/hash scripts unless explicitly authorized.
Kubernetes API	`nmap -sV -p 6443,8443,10250,10257,10259 TARGET -oA outputs/k8s_detect_TARGET`	Detection only.	XML.	Local install did not include a Kubernetes NSE script; verify local script DB.
Docker API	`nmap -sV --script docker-version -p 2375,2376 TARGET -oA outputs/docker_TARGET`	Docker API version if exposed.	XML.	Detection only; exposed Docker API is high impact.
Printers/IoT	`nmap -sV --version-light -p 80,443,515,631,9100 TARGET -oA outputs/printer_TARGET`	Minimal service metadata.	XML.	Fragile devices. Exclude by default unless explicitly in scope.

Web Service Enumeration With Nmap

Recipe	Command	Output	Notes
Common web ports	`nmap -sV -p 80,443,8000,8080,8081,8443,9000,9443 TARGET -oA outputs/web_ports`	Service inventory.	Include app-specific ports from CMDB.
Titles and headers	`nmap -sV --script http-title,http-headers -p 80,443,8080,8443 TARGET -oA outputs/web_meta`	Page title and headers.	Virtual hosts can change content.
Methods check	`nmap --script http-methods -p 80,443 TARGET -oA outputs/http_methods`	Allowed HTTP methods.	Methods are not automatically vulnerabilities.
robots.txt	`nmap --script http-robots.txt -p 80,443 TARGET -oA outputs/http_robots`	Discovery metadata.	Treat as sensitive; not proof of exposure.
TLS cert metadata	`nmap -sV --script ssl-cert -p 443,8443 TARGET -oA outputs/web_certs`	Cert subject, SANs, validity.	Use `-v` if issuer/fingerprints are needed.
TLS ciphers	`nmap -sV --script ssl-enum-ciphers -p 443,8443 TARGET -oA outputs/web_ciphers`	Protocol/cipher inventory.	Can be slower; batch.

NOTE Nmap is excellent for service discovery and metadata. It is not a replacement for an authenticated web scanner, manual app review, or configuration inspection.

Inventory Workflows

Workflow	Command	Artifact
Discovery to live hosts	`nmap -sn -n -iL scope/targets.txt -oA outputs/discovery`	`outputs/discovery.xml`, `scope/live-hosts.txt`
Live hosts to TCP inventory	`sudo nmap -sS --top-ports 1000 --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_top`	Open TCP baseline.
Full TCP after top ports	`sudo nmap -sS -p- --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_all`	Complete listener baseline.
Service inventory	`nmap -sV --version-light -iL scope/live-hosts.txt -p PORTS -oA outputs/services`	Service, product, version, CPE.
CPE inventory	Parse `service/cpe` from XML.	Enrichment input, not proof of vulnerability.
Periodic scans	Same scope, same options, same output naming.	Comparable XML.
Diffing	`ndiff outputs/previous.xml outputs/current.xml > reports/delta.txt`	Change report.
DHCP churn handling	Keep MAC, hostname, IP, and first/last seen where available.	Reduces false "new host" noise.

Inventory Pipeline

```
nmap -sn -n -iL scope/targets.txt --excludefile scope/exclude_fragile.txt -oA outputs/discovery_YYYYMMDD_SCOPE
awk '/Status: Up/{print $2}' outputs/discovery_YYYYMMDD_SCOPE.gnmap | sort -u > scope/live-hosts.txt
sudo nmap -sS --top-ports 1000 --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_top_YYYYMMDD_SCOPE
nmap -sV --version-light -iL scope/live-hosts.txt -p 22,80,443,445,3389 -oA outputs/services_YYYYMMDD_SCOPE
xmlstarlet sel -T -t -m '//host/ports/port[state/@state="open"]' \
  -v 'ancestor::host/address[1]/@addr' -o '\t' -v '@portid' -o '\t' \
  -v 'service/@name' -o '\t' -v 'service/@product' -o '\t' \
  -v 'service/@version' -n outputs/services_YYYYMMDD_SCOPE.xml > reports/services_YYYYMMDD.tsv
```

Review question	Why it matters
New host?	Could be valid provisioning, rogue asset, DHCP churn, or previous false negative.
New port?	Could be release change, misconfiguration, or scan inconsistency.
Service changed?	Version probes and banners can vary; verify before filing.
Missing host?	Could be offline, filtered, retired, or scan failure.
Duplicate hostname?	Common with golden images, stale DNS, or DHCP reuse.

Diffing and Change Detection

Task	Command	Notes
Basic diff	`ndiff outputs/old.xml outputs/new.xml`	Compares Nmap XML.
Save delta	`ndiff outputs/old.xml outputs/new.xml > reports/ndiff_YYYYMMDD.txt`	Attach to weekly review.
Stable comparison	Reuse same target list, probes, timing, and exclusions.	Reduces false deltas.
New hosts	Compare discovery XML or live-host lists.	Confirm with CMDB/DHCP.
Port changes	Diff top/full TCP outputs separately.	Do not compare different scan shapes.
Service changes	Diff service XML from same `-sV` options.	Banners can change without patch changes.

```
ndiff outputs/tcp_top_20260701.xml outputs/tcp_top_20260708.xml > reports/tcp_delta_20260708.txt
awk '/^\+|^-/{print}' reports/tcp_delta_20260708.txt > reports/tcp_delta_summary_20260708.txt
```

False-change source	Control
Firewall transient state	Repeat a small sample and correlate logs.
Packet loss	Slow timing, increase retries, scan from stable path.
DNS churn	Use IP scope and resolve names separately.
Different Nmap versions	Record `nmap --version` in metadata.
Different scan privileges	SYN vs connect scans can differ.

Piping Into and Out of Nmap

Goal	Command	Requirements
Clean scope list	`awk '{print $1}' scope.txt | sort -u > targets.txt`	POSIX `awk`, `sort`.
From JSON inventory	`jq -r '.assets[].ip // empty' assets.json | sort -u > scope/targets.txt`	`jq`.
From notes	`rg -o '([0-9]{1,3}.){3}[0-9]{1,3}' notes/ | awk -F: '{print $2}' | sort -u > scope/targets.txt`	`rg`, POSIX tools.
From git-tracked scope	`git ls-files 'scope/*.txt' | xargs awk '{print $1}' | sort -u > scope/targets.txt`	`git`, `xargs`.
Feed targets	`nmap -sn -n -iL scope/targets.txt -oA outputs/discovery`	Nmap.
Extract up hosts	`awk '/Status: Up/{print $2}' outputs/discovery.gnmap > scope/live-hosts.txt`	Grepable output.
XML open services	Use `xmlstarlet sel` against `outputs/scan.xml`.	`xmlstarlet`.
Tee human output	`nmap -sV TARGET -oA outputs/scan | tee reports/scan_console.txt`	`tee`; still keep `-oA`.

XML to TSV

```
xmlstarlet sel -T -t \
  -m '//host/ports/port[state/@state="open"]' \
  -v 'ancestor::host/address[1]/@addr' -o '\t' \
  -v '@protocol' -o '\t' \
  -v '@portid' -o '\t' \
  -v 'service/@name' -o '\t' \
  -v 'service/@product' -o '\t' \
  -v 'service/@version' -n outputs/scan.xml > reports/open_services.tsv
```

FOOTGUN Avoid brittle parsing of normal output. `-oG` is useful for quick host extraction, but XML should drive durable reporting.

Nmap With Git, Notes, and Reporting Workflows

Practice	Example	Caution
Store scope metadata	`scope/targets.txt`, `scope/exclude_fragile.txt`, `scope/README.md`	Only when policy allows.
Record commands	`printf '%s %sn' "$(date -u +%FT%TZ)" "nmap -sV ..." >> commands.log`	Commands can reveal targets. Protect repo.
Keep outputs out of public repos	Add `outputs/` and `reports/private/` to `.gitignore`.	Scan results are sensitive.
Version report templates	Commit sanitized report skeletons.	Redact IPs, hostnames, banners, cert SANs as needed.
Reproducible scans	Save Nmap version, source IP, scope hash, exclusions, timing.	Reproducibility beats memory.
Delta changelog	`reports/changelog.md` with date, scope, options, summary.	Do not overstate unverified deltas.

Project Layout

```
nmap-project/
  README.md
  commands.log
  scope/
    targets.txt
    exclude_fragile.txt
  outputs/
    20260706_scope_discovery.xml
    20260706_scope_tcp_top.xml
  reports/
    services_20260706.tsv
    ndiff_20260706.txt
```

Scan Metadata Block

```
Scan purpose: authorized internal asset inventory
Scope source: scope/targets.txt
Exclusions: scope/exclude_fragile.txt
Scanner host/source IP: SCANNER_HOST / SCANNER_IP
Nmap version: nmap --version output saved
Command: nmap -sV --version-light --reason -iL scope/live-hosts.txt -oA outputs/services_YYYYMMDD_SCOPE
Timing window: 2026-07-06 22:00-23:30 local
Notification: SOC ticket or change record
Limitations: unauthenticated network view; UDP not exhaustive; banners unverified
```

Troubleshooting Scans

Symptom	Commands	Likely causes
No hosts found	**`nmap -sn -PE -PS22,80,443 -PA80,443 TARGETS --reason`**	Firewall, wrong interface, VPN route, DNS target error.
All ports filtered	**`nmap --reason -p 22,80,443 TARGET`**	Firewall, ACL, host down, wrong source path.
DNS weirdness	**`nmap -n ...`** then **`nmap -R ...`**	Resolver delays, split DNS, stale PTR.
Permission problem	**`nmap -sS TARGET`** then **`sudo nmap -sS TARGET`**	Raw socket privileges.
Raw socket issue	**`nmap --unprivileged -sT TARGET`**	Fall back to connect scan.
VPN/interface issue	**`nmap --iflist`** and **`sudo nmap -e IFACE -sn TARGETS`**	Wrong route or interface.
Local firewall	Test a known lab host and check local host firewall rules.	Scanner-side filtering.
Target firewall	**`nmap --reason -Pn -p PORT TARGET`**	Discovery blocked, port filtered.
Packet loss	Lower rate, raise retries, scan smaller batch.	Congestion or path filtering.
IPv6 surprise	**`nmap -6 -sn -iL scope/ipv6_targets.txt`**	Dual-stack service exposure.
False service detection	**`nmap -sV --version-trace -p PORT TARGET -oN reports/version_trace.txt`**	Proxy, banner deception, TLS tunnel.
UDP ambiguity	**`sudo nmap -sU --reason -p PORT TARGET`**	ICMP rate limiting or silent service.
Slow scans	Add **`--host-timeout`**, lower scope, tune retries.	UDP, filtered ports, retries.
XML parsing error	Validate file path, ensure scan completed, parse XML not normal output.	Interrupted or mixed outputs.

Known-Good Lab Check

```
nmap --iflist
nmap -sn -n LAB_TARGET -oA outputs/lab_discovery_check
nmap --reason -p 22,80,443 LAB_TARGET -oA outputs/lab_port_check
nmap -sV --version-light -p 22,80,443 LAB_TARGET -oA outputs/lab_sv_check
```

NOTE **`--packet-trace`**, **`-d`**, and **`--script-trace`** are diagnostics. Use them narrowly and protect the resulting logs.

Privileges and Platform Notes

Platform/topic	Notes
macOS root/admin	Raw packet operations such as SYN scan, OS detection, and many discovery modes require **`sudo`**. Homebrew Nmap works well but still needs privileges for raw sockets.
Linux root/capabilities	Root is simplest. Some deployments grant capabilities such as raw socket access to the Nmap binary; document this if used.
Windows admin	Administrator privileges and Npcap affect raw packet features. Validate with a small approved target.
SYN fallback	Without privileges, Nmap often falls back to TCP connect behavior where possible. Do not compare privileged and unprivileged results blindly.
Packet capture permissions	If validating with packet capture, capture tooling may need separate privileges.
Interface selection	Use **`nmap --iflist`**, then **`-e IFACE`** when multiple NICs or VPNs exist.
VPN routing	Split tunnels can send scans down the wrong path. Confirm route and source IP first.
IPv6	**`-6`** is explicit. IPv6 scope should be target-list driven, not naive subnet sweeping.
Docker/container limits	Containers may lack raw sockets, host networking, or route visibility. Prefer a controlled scanner host for authoritative scans.
External tools	**`jq`**, **`xmlstarlet`**, **`xsltproc`**, **`ndiff`**, **`rg`**, and GNU utilities may need installation. Label requirements in runbooks.

Defensive Scan Design

Planning item	Field decision
Target scope	Exact CIDRs, hosts, cloud accounts, VLANs, and ownership.
Exclusions	Fragile systems, OT, printers, regulated devices, third parties, maintenance freezes.
Scan purpose	Inventory, change validation, TLS review, exposure check, incident scoping.
Allowed techniques	Discovery, TCP, UDP sample, service detection, safe NSE, named vuln checks.
Timing window	Local time, expected duration, batch size, stop condition.
Rate limits	Max rate, retries, scan delay, parallelism limits.
Output path	**`outputs/`**, **`reports/`**, retention and access controls.
Notification plan	SOC, NOC, app owners, change ticket, scanner source IP.
Rollback/stop condition	Who can stop the scan and what telemetry triggers a stop.
Validation path	How candidate findings become confirmed findings.

Planning Checklist

Item	Done
Scope file reviewed and list scan saved with **`-sL -n`**.	
Exclusion file reviewed with asset owners.	
Scan source IP and interface confirmed.	
SOC/NOC notified with time window and source IP.	
Conservative timing chosen for production.	
UDP and NSE scripts explicitly justified.	
Outputs saved with date/scope basename.	
Stop condition and contact named.	
Results will be validated before ticketing.	
Sensitive outputs protected from public repos and broad sharing.	

WARN Scan results are evidence candidates. Treat open ports, banners, NSE messages, and CPEs as observations that need context, owner review, and validation.

Advanced Combo Patterns

Authorized Subnet Inventory

Element	Details
Problem	Build a repeatable TCP service inventory for an approved subnet.
Command sequence	See below.
Why shaped this way	List scan prevents scope mistakes; discovery reduces full-scan load; version detection runs after port inventory.
Footguns	DHCP churn, missing DNS, packet loss, and unapproved exclusions.

```
nmap -sL -n -iL scope/targets.txt --excludefile scope/exclude_fragile.txt -oN reports/preflight.txt
nmap -sn -n -iL scope/targets.txt --excludefile scope/exclude_fragile.txt -oA outputs/discovery_SCOPE_DATE
awk '/Status: Up/{print $2}' outputs/discovery_SCOPE_DATE.gnmap | sort -u > scope/live-hosts.txt
sudo nmap -sS --top-ports 1000 --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_top_SCOPE_DATE
nmap -sV --version-light -iL scope/live-hosts.txt -p 22,80,443,445,3389 -oA outputs/services_SCOPE_DATE
```

All TCP Ports Only on Confirmed Live Hosts

```
sudo nmap -sS -p- --open --reason --max-retries 3 --host-timeout 45m \
  -iL scope/live-hosts.txt -oA outputs/alltcp_SCOPE_DATE
```

Point	Detail
Why	Reduces noise compared with `-Pn -p-` across the whole scope.
Footguns	Discovery false negatives can exclude live hosts; sample-check filtered segments.

Targeted UDP Validation After TCP Inventory

```
awk '$3 ~ /domain|ntp|snmp|ldap|kerberos/ {print $1}' reports/services.tsv | sort -u > scope/udp-candidates.txt
sudo nmap -sU -sV --version-light -p 53,88,123,161,389 -iL scope/udp-candidates.txt -oA outputs/udp_targeted_SCOPE_DATE
```

Point	Detail
Why	UDP is expensive; candidate-driven scanning lowers impact.
Footguns	TCP inventory does not prove UDP presence; candidate logic is a convenience, not authority.

TLS Certificate Inventory Across Web Hosts

```
nmap -sV --script ssl-cert -p 443,8443,9443 -iL scope/web-hosts.txt -oA outputs/tls_certs_SCOPE_DATE
xmlstarlet sel -T -t -m '//host/ports/port/script[@id="ssl-cert"]' \
  -v 'ancestor::host/address[1]/@addr' -o '\t' \
  -v 'ancestor::port/@portid' -o '\t' \
  -v 'elem[@key="subject"]/table/elem[@key="commonName"]' -n \
  outputs/tls_certs_SCOPE_DATE.xml > reports/tls_subjects_SCOPE_DATE.tsv
```

Point	Detail
Why	Certificate identity review catches wrong certs, expiring certs, and unexpected names.
Footguns	XML shape can vary by script version; verify extraction against sample output.

SSH Algorithm and Hostkey Inventory

```
nmap -sV --script ssh-hostkey,ssh2-enum-algos -p 22 -iL scope/ssh-hosts.txt -oA outputs/ssh_crypto_SCOPE_DATE
```

Point	Detail
Why	Identifies legacy algorithms and duplicated host keys for crypto hygiene review.
Footguns	Do not interpret algorithm presence without policy context and server config review.

SMB Security Mode Inventory

```
nmap -sV --script smb-protocols,smb2-security-mode,smb2-time -p 445 -iL scope/windows-hosts.txt -oA outputs/smb_posture_SCOPE_DATE
```

Point	Detail
Why	Internal Windows posture: dialects, signing, and time skew clues.
Footguns	Domain controllers and file servers are sensitive; avoid broad vuln categories by default.

DNS Service Inventory

```
sudo nmap -sU -sT -p U:53,T:53 --script dns-recursion,dns-service-discovery -iL scope/dns-hosts.txt -oA outputs/dns_SCOPE_DATE
```

Point	Detail
Why	Confirms resolver exposure and DNS-SD metadata in authorized networks.
Footguns	Recursion expected internally can be fine; context determines risk.

Weekly Comparison With ndiff

```
ndiff outputs/tcp_top_LAST.xml outputs/tcp_top_CURRENT.xml > reports/tcp_top_delta_CURRENT.txt
ndiff outputs/services_LAST.xml outputs/services_CURRENT.xml > reports/services_delta_CURRENT.txt
```

Point	Detail
Why	Separates port-state changes from service/version changes.

Point	Detail
Footguns	Different flags or Nmap versions reduce comparison value.

Host:Port:Service TSV From XML

```
xmlstarlet sel -T -t \
  -m '//host/ports/port[state/@state="open"]' \
  -v 'ancestor::host/address[1]/@addr' -o ':' -v '@portid' -o ':' \
  -v '@protocol' -o ':' -v 'service/@name' -n outputs/services.xml \
  > reports/host_port_service.tsv
```

Point	Detail
Why	Produces a compact artifact for triage, diff review, and owner routing.
Footguns	Do not lose scan metadata; TSV alone is not evidence.

Large Scan With Resume-Safe Output

```
sudo nmap -sS -p- --open --reason --max-rate 500 --max-retries 3 --host-timeout 45m \
  -iL scope/live-hosts.txt -oA outputs/alltcp_large_SCOPE_DATE
nmap --resume outputs/alltcp_large_SCOPE_DATE.nmap
```

Point	Detail
Why	Normal output supports resume and XML supports later automation.
Footguns	Keep the original **`.nmap`** file intact. Do not edit options into the resume file.

Cautious Profile for Fragile Networks

```
sudo nmap -sS -p 22,80,443,445,3389 --open --reason \
  --max-rate 50 --scan-delay 100ms --max-retries 5 \
  -iL scope/fragile-approved.txt -oA outputs/fragile_cautious_SCOPE_DATE
```

Point	Detail
Why	Narrow ports, low rate, explicit target list, and clear evidence.
Footguns	Even cautious scans can affect fragile systems; get owner approval and stop criteria.

Lab-Only Deep Scan Profile

```
sudo nmap -sS -sU -A -p T:1-65535,U:53,67,68,69,123,161,500,514,1900 \
  --version-all --reason -iL scope/lab-hosts.txt -oA outputs/lab_deep_SCOPE_DATE
```

Point	Detail
Why	Combines full TCP, selected UDP, OS, version, scripts, and traceroute for controlled lab baselining.
Footguns	Not a production default. **`-A`**, UDP, and all TCP ports are high visibility and higher impact.

Appendix: Compact Reference Tables

Target Syntax

Syntax	Example	Notes
Hostname	`app01.example.test`	DNS controls target identity.
IPv4	`192.0.2.10`	Exact host.
IPv6	`2001:db8::10` with `-6`	Use explicit lists.
CIDR	`192.0.2.0/24`	Confirm ownership.
Range	`192.0.2.10-50`	Lab and small blocks.
Input file	`-iL scope/targets.txt`	Preferred for repeatability.
Exclude	`--exclude 192.0.2.5`	One-off exclusion.
Exclude file	`--excludefile scope/exclude.txt`	Operational default.

Discovery Flags

Flag	Meaning	Notes
`-sn`	Host discovery only	No port scan.
`-sL`	List targets	Preflight scope.
`-Pn`	Treat hosts as online	Narrow use; broad scans become noisy.
`-PE`	ICMP echo	Often filtered.
`-PP`	ICMP timestamp	Often filtered.
`-PM`	ICMP netmask	Often filtered.
`-PS`	TCP SYN discovery	Add ports: `-PS22,80,443`.
`-PA`	TCP ACK discovery	Add ports: `-PA80,443`.
`-PU`	UDP discovery	Add ports: `-PU53,123`.
`-n`	No DNS	Faster and quieter.
`-R`	Always reverse-resolve	Inventory enrichment.

Scan Types

Flag	Type	Privilege
`-sS`	TCP SYN	Root/admin.
`-sT`	TCP connect	Unprivileged.
`-sU`	UDP	Root/admin preferred.
`-sA`	TCP ACK filter mapping	Root/admin.
`-sW`, `-sM`	Window/Maimon	Edge diagnostics.
`-sN`, `-sF`, `-sX`	Null/FIN/Xmas	Historical/edge diagnostics.

Port Selection

Flag	Meaning
`-p 22,80,443`	Explicit ports.
`-p 1-1024`	Range.
`-p-`	All TCP ports when used with TCP scan.
`-p U:53,123,T:22,443`	Mixed UDP/TCP list.
`--top-ports 1000`	Most common ports.
`--port-ratio RATIO`	Ports above frequency ratio.
`-F`	Fast mode, fewer ports.
`--exclude-ports PORTS`	Exclude specified ports.
`-r`	Sequential order, no randomization.

Service and OS Flags

Flag	Meaning
`-sV`	Service/version detection.
`--version-light`	Lower-intensity probes.
`--version-intensity 0-9`	Tune version probe intensity.
`--version-all`	Try all probes.
`--version-trace`	Debug version detection.
`-O`	OS detection.
`--osscan-limit`	Only promising targets.
`--osscan-guess`	More aggressive guessing.
`-A`	OS, version, default scripts, traceroute.

Timing and Performance

Flag	Meaning
`-T0` to `-T5`	Timing templates, slower to faster.
`--min-rate N`	Send no slower than N packets/second.
`--max-rate N`	Send no faster than N packets/second.
`--max-retries N`	Cap retransmissions.
`--host-timeout TIME`	Give up on slow host.
`--scan-delay TIME`	Delay between probes.
`--max-scan-delay TIME`	Cap adaptive scan delay.
`--min-parallelism N`	Minimum probe parallelism.
`--max-parallelism N`	Maximum probe parallelism.

NSE Categories

Category	Default stance
`default`	Useful baseline with permission.
`safe`	Preferred for routine discovery.
`discovery`	Inventory metadata.
`version`	Version-detection helper scripts.
`vuln`	Narrow validation.
`auth`	Avoid unless explicitly authorized.
`brute`	Avoid by default.
`exploit`	Lab/explicit validation only.
`intrusive`	Avoid by default.
`dos`	Avoid.
`external`	Privacy review required.

Output and Troubleshooting Flags

Flag	Meaning
`-oN FILE`	Normal output.
`-oX FILE`	XML output.
`-oG FILE`	Grepable output.
`-oA BASENAME`	Normal, XML, grepable.
`--append-output`	Append instead of overwrite.
`--resume FILE`	Resume interrupted scan.
`--reason`	State reason.
`--open`	Show open or possibly open.
`-v`, `-vv`	Verbosity.
`-d`, `-dd`	Debugging.
`--packet-trace`	Packet-level debug.
`--iflist`	Interfaces and routes.
`--traceroute`	Path clue after scan.

Common Ports by Protocol

Protocol	Ports
SSH	TCP 22
HTTP/HTTPS	TCP 80, 443, 8080, 8443, 8000, 9443
DNS	UDP/TCP 53
SMTP	TCP 25, 465, 587
SMB	TCP 445, 139
RDP	TCP 3389
LDAP/LDAPS	TCP/UDP 389, TCP 636
Kerberos	TCP/UDP 88, TCP/UDP 464
SNMP	UDP 161, 162
NTP	UDP 123
PostgreSQL	TCP 5432
MySQL	TCP 3306
MSSQL	TCP 1433, UDP 1434
Docker API	TCP 2375, 2376
Kubernetes API and components	TCP 6443, 8443, 10250, 10257, 10259
Printing	TCP 515, 631, 9100

False Positives and False Negatives

Pattern	Likely cause	Response
Host absent in discovery	Filtered probes or offline host	Repeat with mixed probes and narrow **`-Pn`**.
Port **`filtered`**	Firewall/ACL/drop behavior	Use **`--reason`**, validate path, coordinate with owners.
UDP **`open|filtered`**	Silent UDP or ICMP rate limit	Treat as ambiguous; use targeted version checks.
Wrong service	Proxy, TLS, banner deception	Use **`-sV --version-trace`** on one port.
Wrong OS	NAT, firewall, virtualization	Report as clue only.
New port in diff	Real change or scan inconsistency	Re-run same command against sample and check logs.
Missing service in diff	Packet loss, host offline, ACL	Repeat with slower timing.

Safe Default Profiles

Profile	Command
Scope preflight	`nmap -sL -n -iL scope/targets.txt --excludefile scope/exclude_fragile.txt -oN reports/preflight.txt`
Discovery	`nmap -sn -n -iL scope/targets.txt --excludefile scope/exclude_fragile.txt -oA outputs/discovery_SCOPE_DATE`
TCP top	`sudo nmap -sS --top-ports 1000 --open --reason -iL scope/live-hosts.txt -oA outputs/tcp_top_SCOPE_DATE`
Service light	`nmap -sV --version-light --reason -p PORTS -iL scope/live-hosts.txt -oA outputs/services_SCOPE_DATE`
Web metadata	`nmap -sV --script http-title,http-headers,http-methods -p 80,443,8080,8443 -iL scope/web-hosts.txt -oA outputs/web_SCOPE_DATE`
TLS metadata	`nmap -sV --script ssl-cert,ssl-enum-ciphers -p 443,8443 -iL scope/web-hosts.txt -oA outputs/tls_SCOPE_DATE`
SSH crypto	`nmap -sV --script ssh-hostkey,ssh2-enum-algos -p 22 -iL scope/ssh-hosts.txt -oA outputs/ssh_SCOPE_DATE`
Diff	`ndiff outputs/previous.xml outputs/current.xml > reports/delta_SCOPE_DATE.txt`

Primary References

Reference	URL
Nmap Reference Guide	`https://nmap.org/book/man.html`
Nmap Scripting Engine usage	`https://nmap.org/book/nse-usage.html`
NSE script documentation	`https://nmap.org/nsedoc/`
Nmap output formats	`https://nmap.org/book/output-formats-commandline-flags.html`
Local verification	`nmap --version`, `nmap -h`, and `nmap --script-help` output from Nmap 7.99 on 2026-07-06